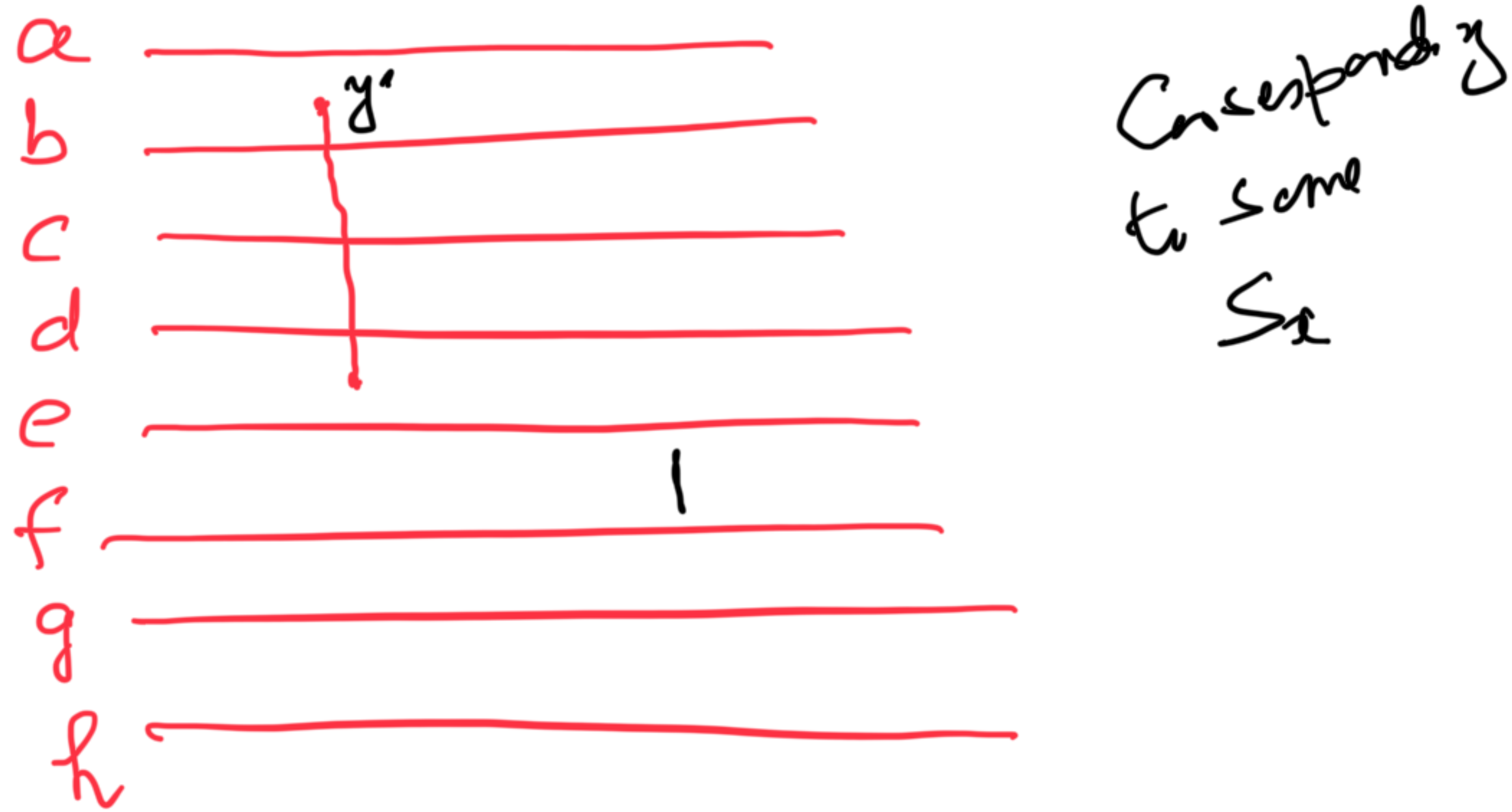
Lecture 3 DAA Jan 28

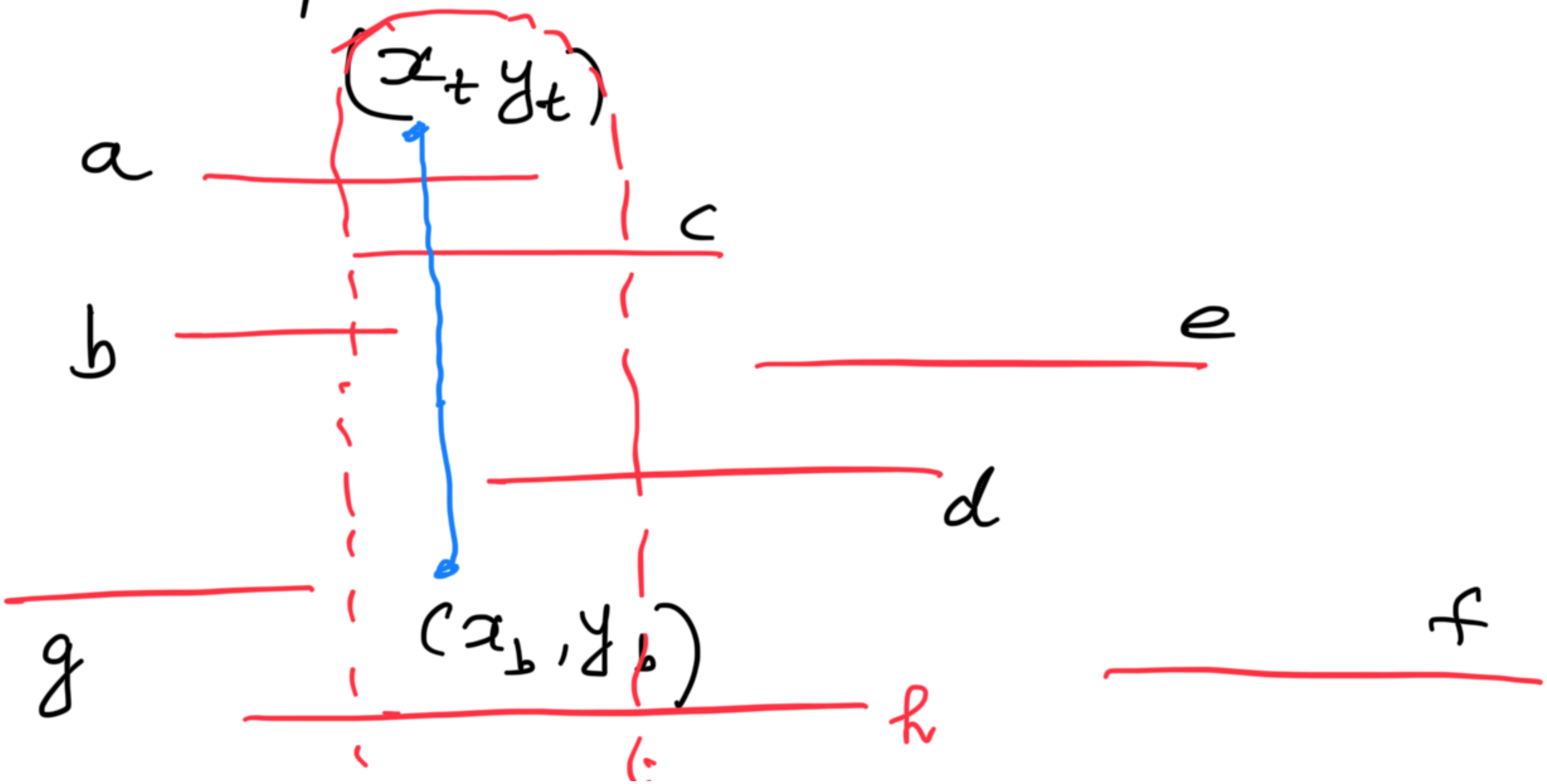


By using binary search to find the first intersection, we can find ~~all~~ the intersections in $O(\log n + k_1)$ steps

intersections

<u>Spl Case</u> : the horizontal segments span the vertical segment

<u>For further thoughts</u> : Given a set of n numbers, can you find if they are all unique



Let the end points of the horizontal segments be denoted as "events" where the sorted set (intersecting with the imaginary vertical line) change

Denote the sorted set of horizontal line segments at any x-coordinate as

$$S_x = \{ \text{set of line segments spanning } x \text{ and sorted as per } y \}$$

Observation : There are only $2 \# N_h$

event points   # horizontal segments

so there are that many distinct sorted sets S_x

$S_1 \quad S_2 \quad S_3 \quad \cdots\cdots\cdots \quad S_{2n}$

in between the sorted set doesn't change

So if the vertical segment V_x occurs between S_i and S_{i+1}, we will use the sorted set S_i

<u>Claim</u> Active sorted set is the only one that that we need to keep —
we can discard the previous ones and compute the the future ones when we encounter the next event point.

The event queue consists of the sorted end points of the horizontal segments as well as the vertical segments — sorted as per x-coordinate —

How do we update S_{i+1} given S_i?

Use any Balanced BST like AVL that has $O(\log n)$ time insert, delete search (for the actual intersections)

Analysis

n = #segments (Vert + hori...)

The event queue has $\leq 2n$ points

$\Rightarrow$ Sorting takes $O(n \log n)$

Work done during sweeping

1. S_i has to be updated (insert a new segment or discard)

   $O(\log n)$ $\Rightarrow$ $O(n \log n)$ for all such events

2. We hit a vertical segment v and all the $h(v)$ intersections are computed in $O(\log n + h(v))$

Total work for all vertical segments

$$= \sum_{\substack{v \in \text{vertical} \\ \text{segments}}} O(\log n + h(v))$$

$$\leq O(n \log n + h)$$

Overall $O(n \log n + h)$

Line sweep framework

What if line segments are not orthonormal